Aadhaar Enrolment Analysis & Societal Trends Intelligence

Project Details

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Team ID: UIDAI_8056

Hackathon: UIDAI Data Hackathon – 2026

Project Title: Aadhaar Enrolment Analysis & Societal Trends Intelligence

Institution / Program: Baba Farid College Of Engineering And Technology /B.Tech (Artificial Intelligence & Machine Learning)

Tools & Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, Power BI, Jupyter Notebook, VS Code

Date of Submission: 20/01/2026

Introduction

This project presents a comprehensive, end-to-end analytical framework for evaluating Aadhaar enrolment trends across India using structured data pipelines, statistical analysis, geospatial intelligence, and interactive visualization. By integrating Excel-based validation, Python-driven exploratory and statistical analytics, PostgreSQL-based database management, and Power BI dashboards, the system transforms raw enrolment records into actionable insights for policymakers and operational stakeholders. The study identifies regional disparities, demographic participation gaps, and temporal trends, enabling data-driven decision-making for improved service accessibility and resource allocation.

Problem Statement

Despite Aadhaar being the world's largest biometric ID system with over 1.3 billion enrolments, significant challenges persist in achieving universal coverage, particularly among vulnerable populations including children (0–17 years), elderly citizens, and residents of geographically remote areas. While child enrolment constitutes a major share of total registrations, substantial disparities exist across states, districts, and demographic segments.

Core Problems Identified

- **Geographic Disparities:** Variations in enrolment efficiency between states, with some achieving over 99% child enrolment while others remain below 90%.
- **Operational Inefficiencies:** Wide variation in "enrolments per district" indicating uneven distribution of enrolment resources.
- **Temporal Inconsistencies:** Weekly and seasonal fluctuations affecting enrolment planning and staffing.

- **Demographic Gaps:** Lower enrolment rates in specific age groups, particularly 0–5 years.
- **Data Quality Issues:** Inconsistent reporting formats and missing geographical granularity in certain regions.

Dataset Used

This project utilizes the Aadhaar Enrolment Dataset provided by the Unique Identification Authority of India (UIDAI) as part of the UIDAI Data Hackathon – 2026. The dataset contains anonymized and aggregated records of Aadhaar enrolments across different geographical regions in India, categorized by age groups and administrative location identifiers.

Scope and Coverage

- **Geographical Coverage:** Pan-India
- **Administrative Levels:** State, District, and PIN Code level
- **Time Dimension:** Daily or periodic records

Dataset Structure

Each row in the dataset represents aggregated enrolment data for a specific date and geographical unit.

Column Name	Description
Date	Date of Aadhaar enrolment record
State	Name of the State or Union Territory
District	Name of the District
PIN Code	Postal Index Number of enrolment locality
Age 0–5	Enrolments for children aged 0–5
Age 5–17	Enrolments for individuals aged 5–17
Age >18	Enrolments for individuals aged above 18

Methodology & Approach

This project follows a structured, multi-phase data analytics and visualization pipeline to ensure accuracy, analytical depth, and decision-oriented outputs.

Phase 1: Dataset Understanding and Excel-Based Inspection

The initial phase focused on dataset familiarization and data quality validation using Microsoft Excel.

Key Activities:

- Verified column consistency across all records
- Identified missing and inconsistent geographical and demographic entries
- Removed duplicate records based on Date–PIN Code combinations
- Performed basic descriptive statistics for each age group
- Validated numerical ranges to ensure logical data integrity

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Phase 2: Data Cleaning and Preprocessing Using Python

The dataset was imported into a Python environment using Pandas and NumPy for scalable processing.

Steps Performed:

- Converted Date column to standardized datetime format
- Handled missing values using conditional imputation and filtering
- Normalized state and district name formats
- Created a derived field, **Total Enrolments**, by summing all age groups
- Removed outliers using IQR and logical threshold checks

Phase 3: Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns, trends, and anomalies.

Techniques Applied:

- Distribution analysis using histograms and box plots
- Time-series analysis for seasonal and growth trends
- Correlation analysis between geography and enrolment volume
- State-wise and district-wise performance benchmarking

Phase 4: Geographical and Demographic Analysis

Geographical Analysis:

- Aggregated data at State, District, and PIN Code levels
- Identified high-density and low-coverage regions
- Mapped regional disparities

Demographic Analysis:

- Computed percentage contribution of each age group
- Compared youth and adult enrolment patterns
- Identified districts with low child and youth participation

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Phase 5: Statistical Analysis and Insight Generation

Methods Used:

- Descriptive statistics (mean, median, variance, standard deviation)
- Growth rate calculations
- Regional performance indexing
- Trend significance testing

Phase 6: Database Integration Using PostgreSQL

To enable scalable querying and structured data management, the cleaned dataset was migrated into a PostgreSQL relational database.

Implementation Steps:

- Designed a normalized relational schema

- Imported data using CSV loaders and batch insertion
- Created indexes on State, District, PIN Code, and Date columns
- Developed optimized SQL queries for:
    - Region-wise enrolment ranking
    - Age-group trend analysis
    - Time-based growth comparisons
    - High-activity PIN Code identification

Phase 7: Power BI Dashboard Development

The final phase focused on developing an interactive visualization system using Microsoft Power BI.

Dashboard Features:

- Live connection to PostgreSQL for dynamic data refresh
- KPI cards for total enrolments and age-group distribution
- Geographical heat maps for regional performance
- Time-series trend charts
- Interactive slicers and drill-through functionality

System Workflow

1. Input dataset is loaded into the system.
2. Data is cleaned, standardized, and validated.
3. Derived metrics and KPIs are computed.
4. Aggregations are performed at state, district, and temporal levels.
5. Visualizations are generated to present insights.
6. Results are interpreted for policy and operational recommendations.

Data Analysis & Visualization Phase

4.1 Descriptive Analysis

- Computed total, mean, median, and standard deviation for each age group
- Compared State-wise and District-wise enrolment volumes
- Evaluated daily and monthly enrolment averages

4.2 Temporal Trend Analysis

- Aggregated data by date and month
- Calculated growth rates and percentage changes
- Identified peak and low-activity periods

4.3 Geographical Analysis

- Ranked regions based on enrolment density
- Identified clusters with low service coverage

4.4 Demographic Analysis

- Analyzed age-group proportional contribution
- Compared regional youth and adult participation

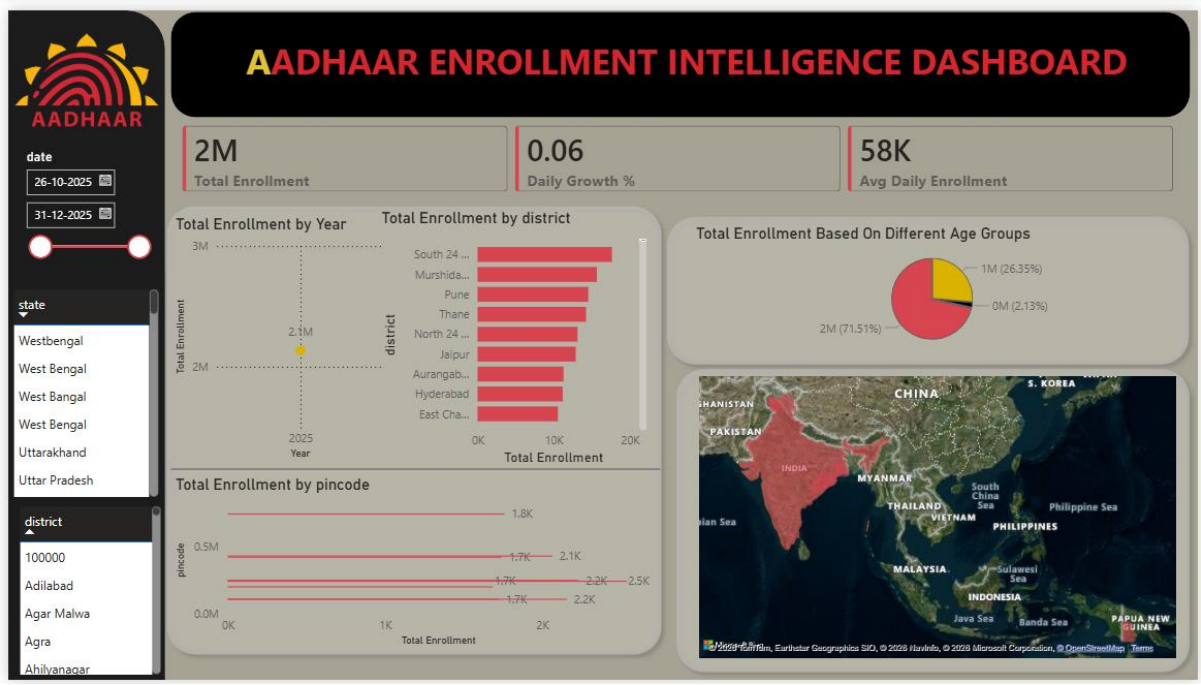
4.5 Statistical and Comparative Analysis

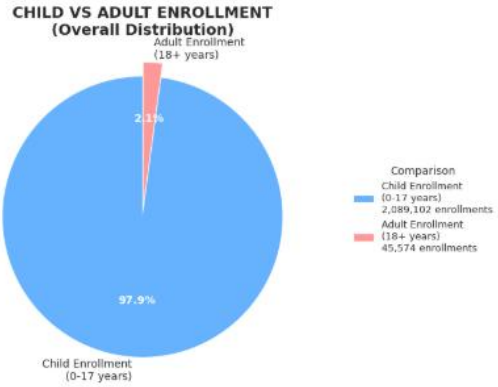
- Normalized data for fair regional comparison
- Calculated variance and correlation metrics

Key Insights Generated

- Identification of states with the highest child enrolment
- Detection of seasonal and peak enrolment periods
- Recognition of districts with high operational efficiency
- Understanding demographic distribution across age groups
- Detection of abnormal growth patterns and anomalies

Project Review



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DATA QUALITY & KEY METRICS
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 OVERALL STATISTICS:
• Total Records: 506,029
• Total Enrollment: 2,134,676
• Child Enrollment: 2,089,102 (97.9%)
• Adult Enrollment: 45,574 (2.1%)

 DATA COMPLETENESS:
• Records with 0-5 data: 450,752 (89.1%)
• Records with 5-17 data: 210,922 (41.7%)
• Records with 18+ data: 21,164 (4.2%)

 STATISTICAL QUALITY:
• Child enrollment outliers: 35,752 (7.1%)
• Skewness: 6.01
• Mean child enrollment: 4.13
• Median child enrollment: 2.00

 GEOGRAPHIC COVERAGE:
• States Covered: 52
• Districts Covered: 1,021
• Avg districts per state: 19.6
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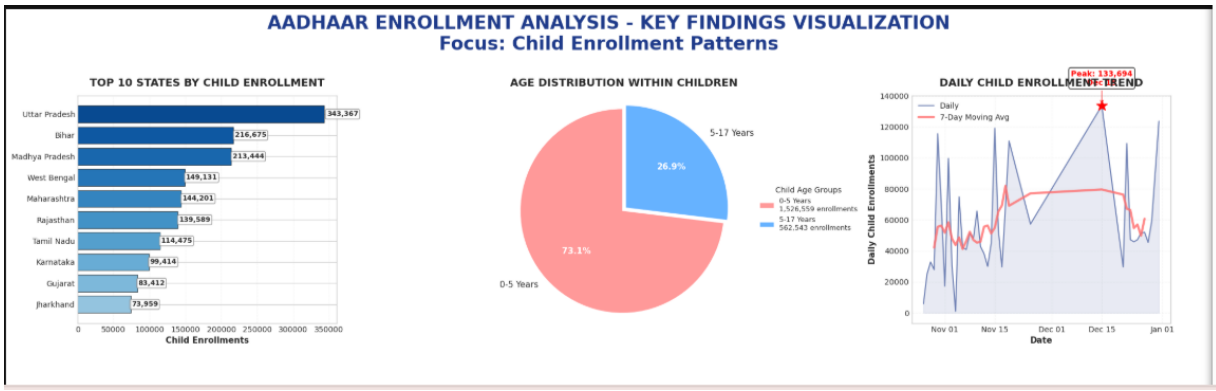

Tools & Technologies Used

- **Python** – Data processing and analytics
- **Pandas & NumPy** – Data manipulation and numerical computation
- **Matplotlib & Seaborn** – Data visualization
- **PostgreSQL** – Relational database management and SQL analytics
- **Power BI** – Interactive dashboards and KPI reporting
- **Jupyter Notebook / VS Code** – Development environment

Conclusion

This project successfully transforms raw Aadhaar enrolment data into a structured intelligence and visualization system that highlights societal trends, regional disparities, and operational performance. The integration of statistical analysis, PostgreSQL-based data management, and Power BI-driven dashboards ensures transparency, improves service delivery, and empowers UIDAI stakeholders with actionable, data-driven insights for sustainable system enhancement.